

Mobile Devices and Applications

UNIT 1 – ANDROID BASICS

Agenda

Objectives

Content

Evaluation

Resources

Objectives

Efficient usage of mobile technologies in the information society context.

Understanding the principles of mobile devices.

Learning and achieving Android development skills.

Evaluation

Course activity: 50%

- Final quiz exam – 50%

Laboratory activity: 50%

- 2 Partial Evaluations – 30%
- Ongoing evaluation – 20%

Final evaluation:

- **Min. 70% laboratory attendance**
- **Min. 2,5 / 5 points for attending the final exam**

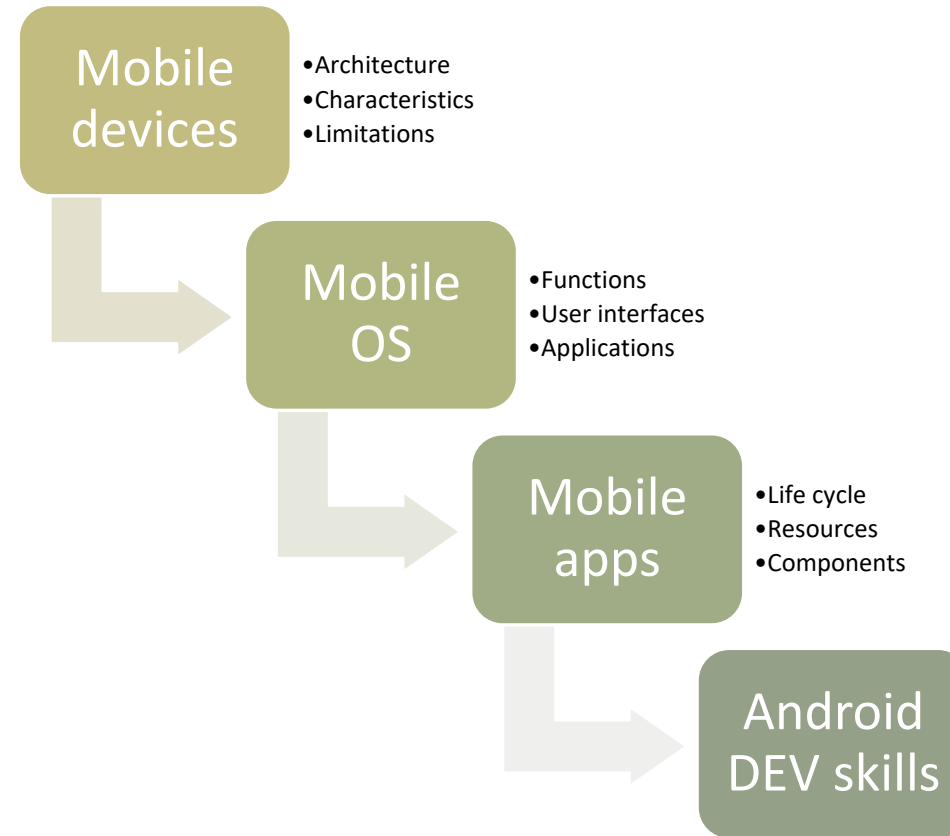
All evaluation activities must be passed with a minimum of 50% from the total number of points

<http://pdm.ase.ro>

Agenda

1. Mobile devices
2. Android framework
3. Development instruments
4. Android application structure

Learning path



Mobile devices

CHARACTERISTICS AND FUNCTIONALITY

Mobile devices

Social networking

Managing personal content (contacts, agenda, calendar)

Internet browsing

Entertainment & Multimedia

- Games
- Virtual Books
- Movies, Images and Music

Communication (e-mails, messages and voice services)

Navigation (GPS, localization)

Financial transactions

Characteristics of Mobile Devices

Portability

Usability

Performance

Communication

Security

Mobile devices limitations

Dimensions:

- Screen, keyboard;

Autonomy (battery life)

Performance

Memory (RAM and ROM)

Bandwidth

User Experience

Form factor

Mobile devices - trends

Fast and reach full development

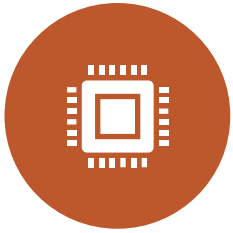
Prices evolution

Hardware and software
performances

Increased usability

Mobile devices

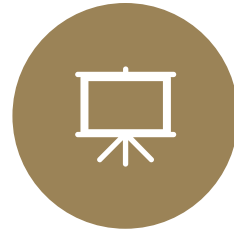
Hardware characteristics



PROCESSOR



MEMORY



SCREEN

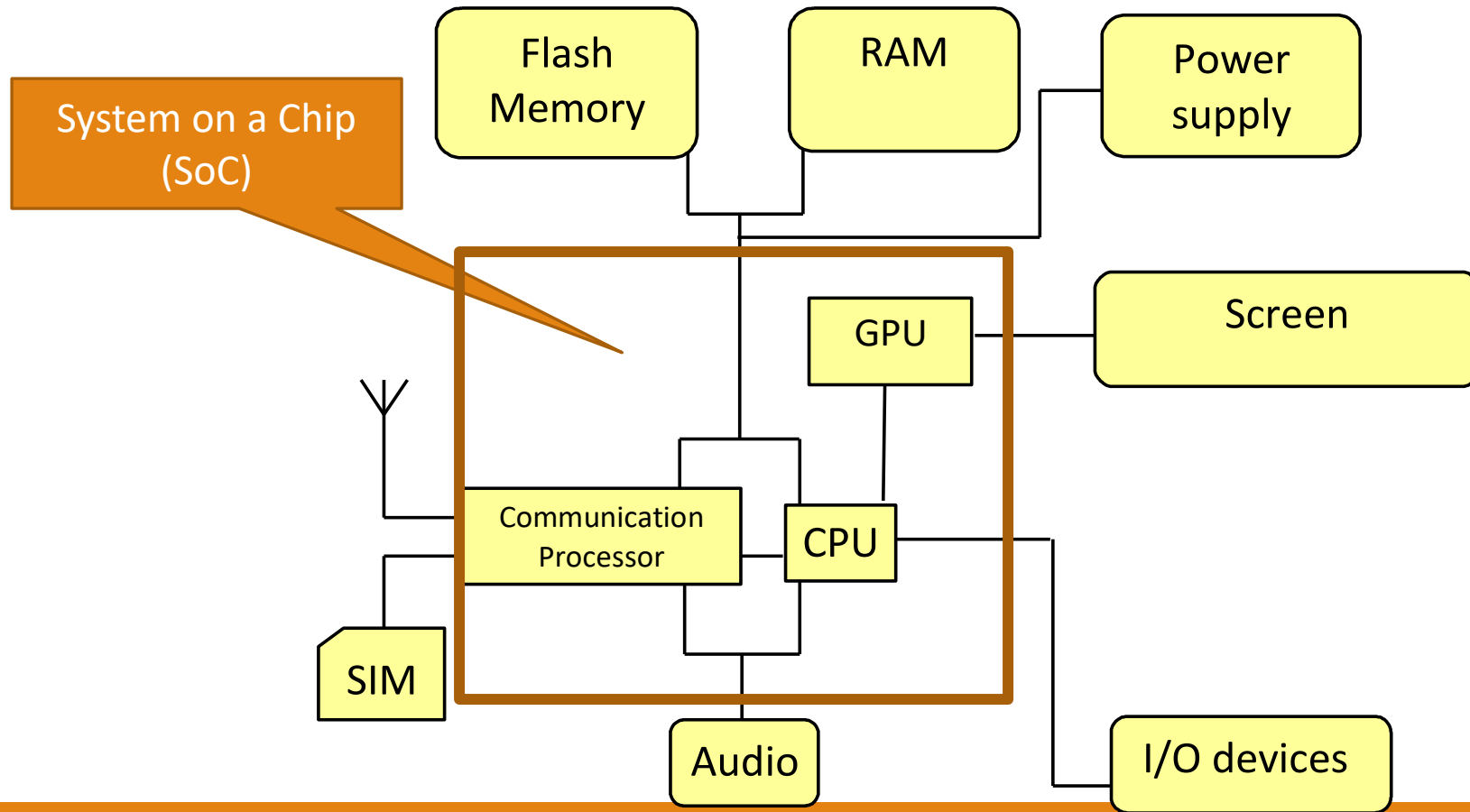


DATA
GATHERING



CONNECTIVITY

Smartphone



SoC

Application processor (CPU)

Memory interface

Graphic processor (GPU)

USB controller

Serial interface

Bluetooth controller

Wi-Fi controller

Camera interface

Application processor

RISC architecture

ARM and Intel

32/64 bits

Working frequencies: around 2.7 GHz

One, dual, quad or octa-core processor

Low power consumption

SoC

Marvell: PXA

Texas
Instruments:
OMAP

Samsung: S3C,
S5C, S5P

Freescale: iMX

Qualcomm:
MSM, QSD

nVidia: Tegra

SoC

Qualcomm

- Snapdragon 400 – 1.6-1.7 GHz, Dual/Quad Core (Nokia Lumia 640/XL, Asus Zenwatch 2, Moto 360 gen. 2, LG Watch Urbane)
- Snapdragon 800 MSM8974A (Google Nexus 5, Galaxy S4)
- Snapdragon 805 APQ8084 (Galaxy Note 4)
- Snapdragon 810 Cortex MSM8x74 (HTC One M9, LG G Flex2, Galaxy S5)
- Snapdragon 835 Kryo (2,45 GHz)
- **Snapdragon 765G with Kryo 475(current Google Pixel 5 specs) (2.45 GHz)**
- **Snapdragon 8 Gen ½ - 4nm – Samsung S22, Xiaomi, Vivo, Motorola, OnePlus**

Samsung

- **Exynos 5 1.9 MHz** (Galaxy Note 3)
- **Exynos 5 Octa** (Galaxy Tab S, Galaxy Alpha, Galaxy A7/A9)
- **Exynos 7 Octa** (Galaxy S6/Edge, Galaxy Note 5)
- **Exynos 2200** 1octa-core Cortex CPU – Samsung Galaxy S22 (for Europe only)

NVIDIA

- **Tegra 4 – 1.7- 1.9 GHz** (NVIDIA Tegra Note, Microsoft Surface 2)
- **Tegra K1**

SoC

Texas Instruments

- **OMAP 4470** (Kindle Fire HD, BN NOKK HD, BlackBerry Z10)

Intel

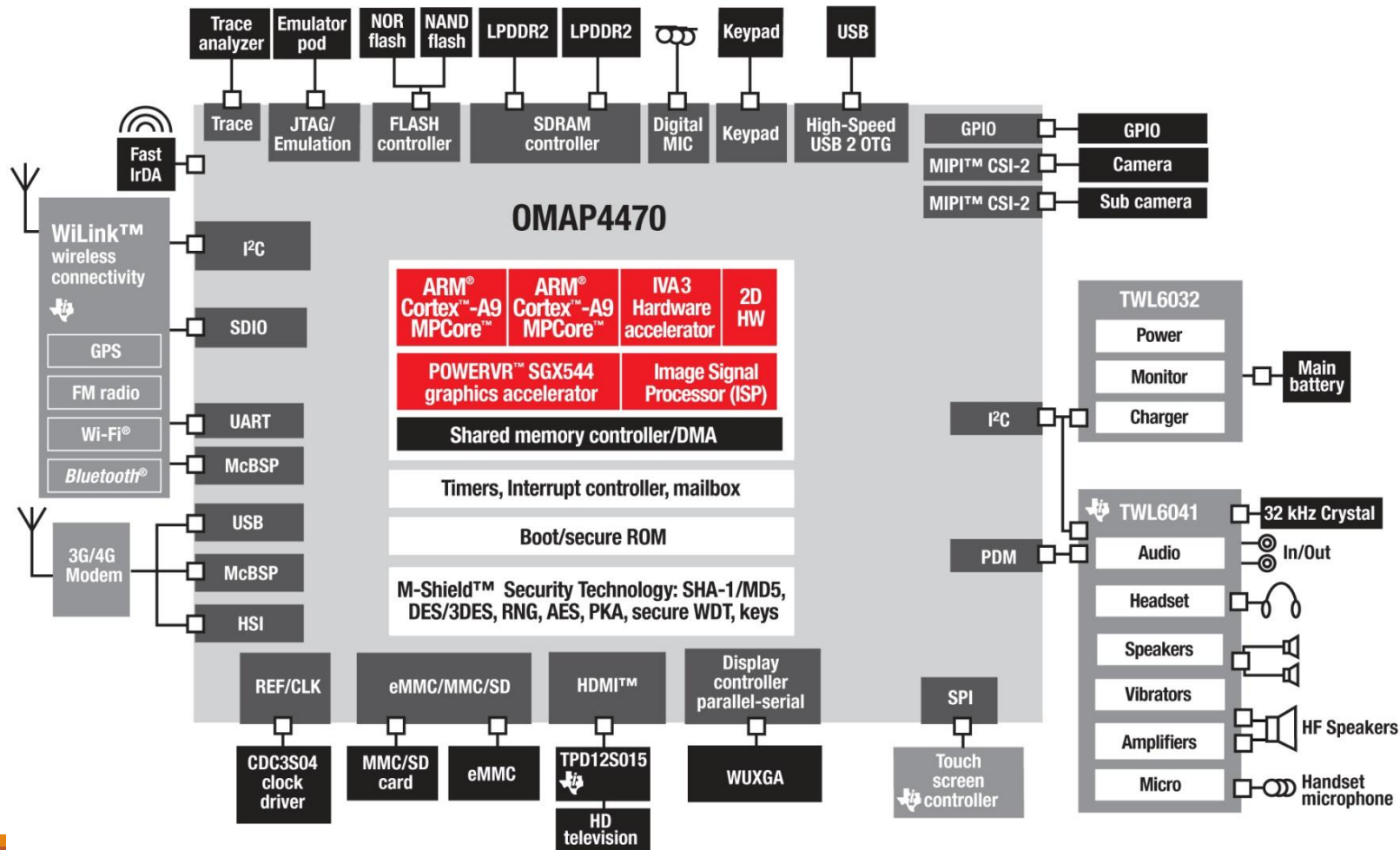
- Atom Z2560 (1.6 GHz) , Z2650 (2 GHz) - (Asus, Lenovo, Dell, Samsung)
- Atom Z3745 1.3 – 1.7 GHz
- Atom C2758, 2.4 GHz
- Atom C5000, 2.8 GHz

Apple

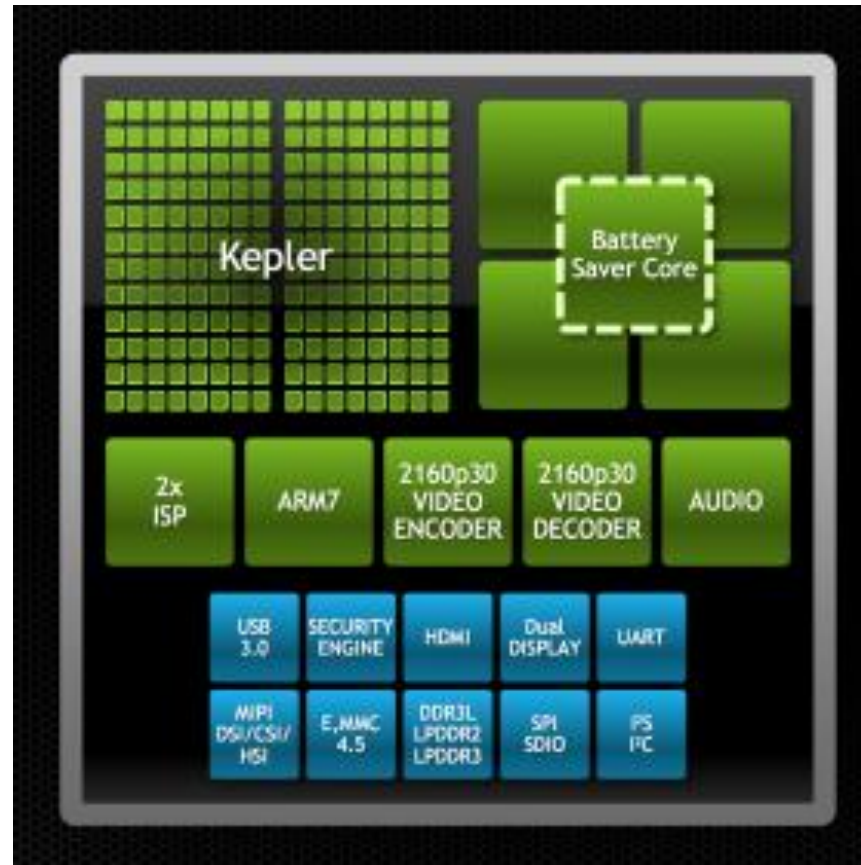
- A5 (iPhone 4S)
- A6 (iPhone 5, 5C)
- A7 (iPhone 5s)
- A8 (iPhone 6)
- A9 (iPhone 6s)
- A10 (iPhone 7)
- A12 (iPhone Xs) – 8 cores CPU, 4 cores GPU
- A12Z -2.65 GHz – instruction set A64 – ARM
- A14 Bionic – iPhone 12
- A15 Bionic – iPhone 13
- A16 Bionic – iPhone 14; (16 billion transistors); 6core CPU, 5nm – 3.46 GHz

Texas Instruments

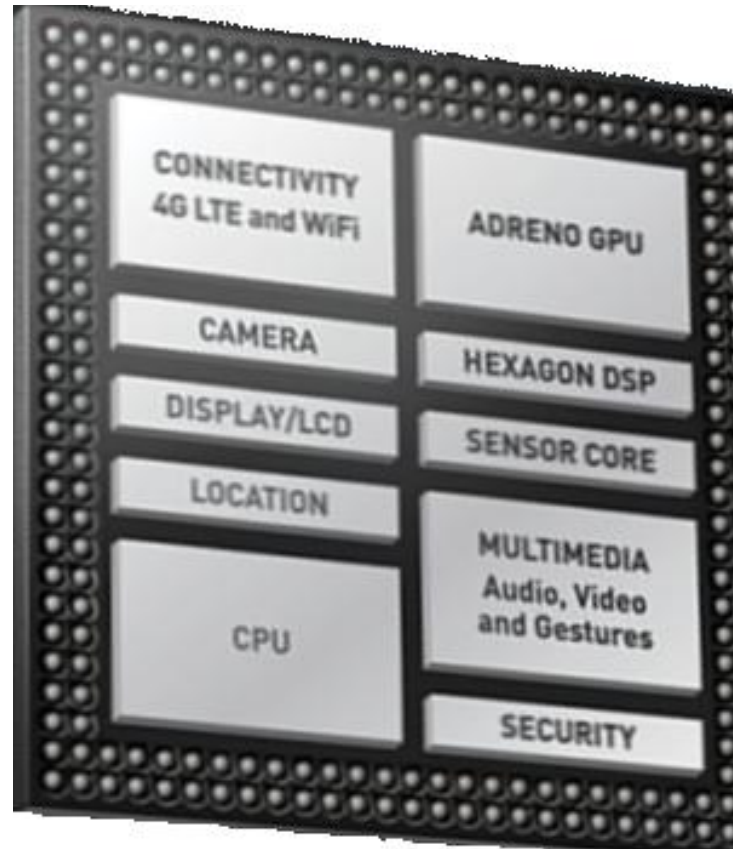
OMAP 44xx



NVIDIA Tegra K1



Qualcomm Snapdragon 80x



Memory

Random Access Memory – RAM

- Up to 6GB of RAM

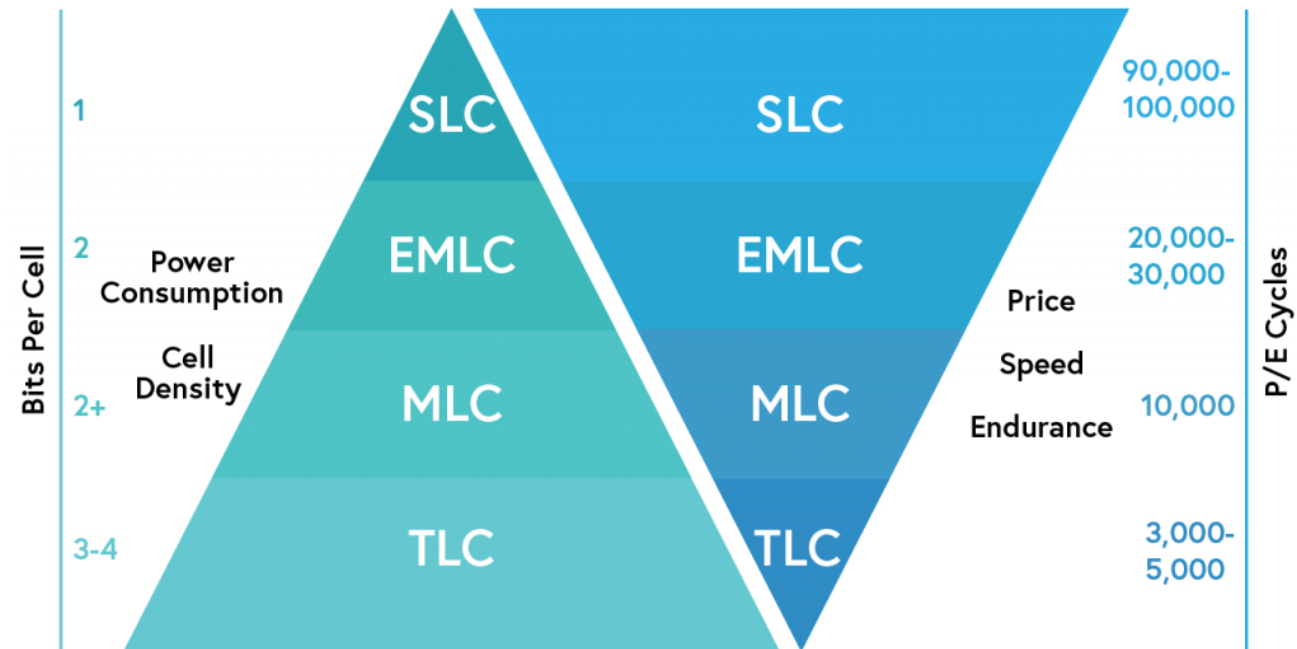
Internal non-volatile memory

- Flash
 - NOR – faster to read than to write, code execution, expensive
 - NAND – cheap access to data storage

External memory

- SD cards

NAND Flash Specs



	SLC	MLC	TLC
Bits per cell	1	2	3
P/E Cycles	100,000	3,000	1,000
Read Time	25 μ s	50 μ s	~75 μ s
Program Time	200-300 μ s	600-900 μ s	~900-1350 μ s
Erase Time	1.5-2 ms	3 ms	4.5 ms



Source: <http://www.anandtech.com/show/6337/samsung-ssd-840-250gb-review/2>

Screen

Diagonal – 6 inches ~ 87.6 cm²

Resolution - 1080 x 2340 pixels

Pixel density (ppi, dpi) ~ 432 ppi density with 19.5:9 ratio

Number of colours - *16M colors*

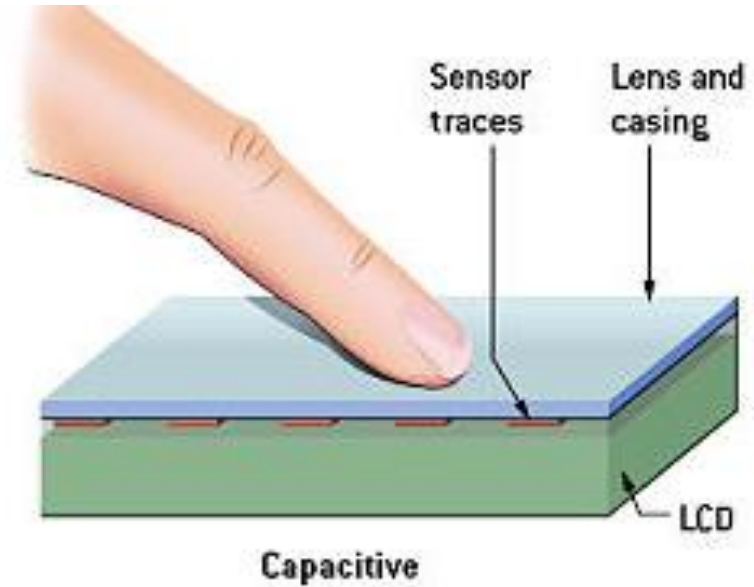
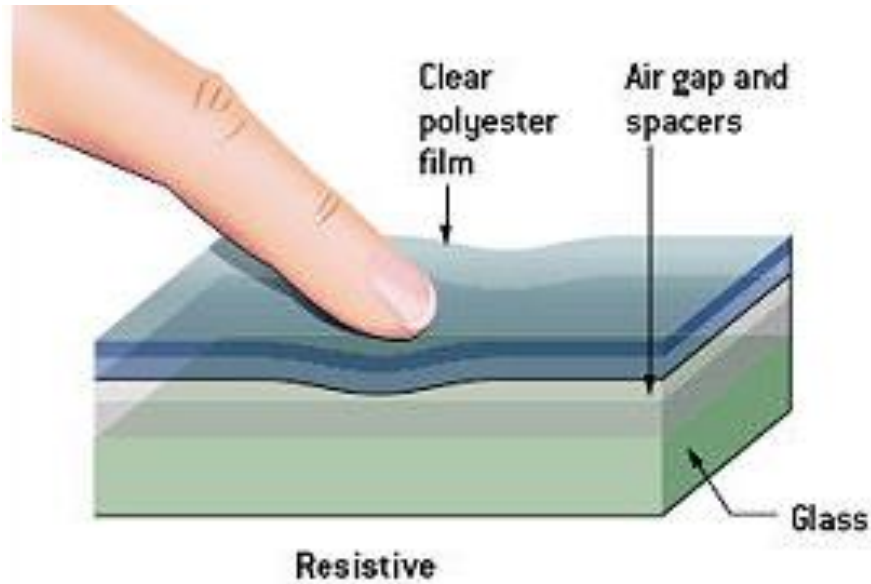
Technology

- LCD TFT (*Thin Film Transistor*)
- LCD IPS (*In Place Switching*)
- OLED (*Organic Light Emitting Diode*) – *Capacitive*
- AMOLED (*Active-Matrix OLED*)
- *Retina*
- *Gorilla Glass*

Touchscreen

- Resistive
- Capacitive

Resistive / Capacitive



Data Input solutions

Touchscreens

- virtual keyboard;

Numeric keyboard

Mini-Joystick (D-pad)

QWERTY keyboard

Techniques:

- Swiping (gestures)
- Vocal commands
- Fingerprint
- Face recognition

Connectivity

Voice services

Data connection

Area coverage

Small – NFC /
Bluetooth

Extended - Wi-Fi
802.11
a/b/g/n/ac

Wireless connectivity

Wireless network	Bandwidth	Technologies
WPAN	9.6 Kbps - 3 Mbps	IR, Bluetooth, NFC
WLAN	1-300 Mbps	802.11 a, b, g, n
WWAN	8 Kbps-300 Mbps	GSM (CSD, HCSD), GPRS, EDGE, UMTS (WCDMA) cu HSPDA/HSPA+, LTE cdmaOne, CDMA2000, 5G

WWAN (GSM)

CSD – Circuit-Switched Data

- **HSCSD** – High Speed Circuit-Switched Data

GPRS – General Packet Radio Services

EDGE – Enhanced Data Rates for Global Evolution

UMTS – Universal Mobile Telecommunications System

- **HSDPA** – High Speed Downlink Packet Access
- **HSUPA** – High Speed Uplink Packet Access

LTE – Long Term Evolution

Mobile Radio Standards

1G

- Radio signals using analog-based protocols (1980s)

2G

- GSM / 2.5 – GPRS, EDGE (1990s) – Digital Cellular

3G

- UMTS, WCDMA, HSPA (2000) – Mobile Broadband

4G

- LTE (2010) – All-IP Broadband

5G

- 5G (2020) – Ultra-Broadband / Low-Latency

Generation	Main Standards	Access Technology	Typical Frequency Bands	Max Data Rate	Typical Latency	Key Features / Notes
1G	AMPS, NMT, TACS	FDMA (Analog)	800–900 MHz	~2.4 kbps (voice only)	> 500 ms	Analog voice, poor security, no data
2G	GSM, IS-95 (cdmaOne), D-AMPS	TDMA / CDMA	850 / 900 / 1800 / 1900 MHz	Up to 64 kbps	~300–500 ms	Digital voice, SMS, basic data
2.5G	GPRS, EDGE	TDMA (Enhanced GSM)	850 / 900 / 1800 / 1900 MHz	GPRS: 114 kbps EDGE: 384 kbps	~200–400 ms	Always-on packet data, MMS
3G	UMTS (W-CDMA), CDMA2000	CDMA (Wideband)	850 / 900 / 1700 / 1900 / 2100 MHz	384 kbps – 2 Mbps	~100–200 ms	Mobile Internet, video calls
3.5G / 3.75G	HSDPA, HSUPA, HSPA+	CDMA (Enhanced W-CDMA)	Same as 3G	Up to 42 Mbps	~50–100 ms	Fast app connectivity, streaming
4G	LTE, LTE-Advanced	OFDMA / SC-FDMA	700 / 800 / 1800 / 2100 / 2600 MHz	100 Mbps – 1 Gbps	~30–50 ms	All-IP, VoLTE, HD video, IoT support
5G	5G NR (Sub-6 & mmWave)	OFDMA (Advanced)	Sub-6 GHz (3.5 GHz) mmWave (24–100 GHz)	Up to 10 Gbps	< 1 ms	Ultra-low latency, massive IoT, network slicing, AR/VR

Resources

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<http://developer.android.com>